

YENJOYY !!

Profit and Loss - Formulas

Cost Price:

The price, at which an article is purchased, is called its cost price, abbreviated as C.P.

Selling Price:

The price, at which an article is sold, is called its selling price, abbreviated as S.P.

Profit or Gain:

If S.P. is greater than C.P., the seller is said to have a profit or gain.

Loss:

If S.P. is less than C.P., the seller is said to have incurred a loss.

IMPORTANT FORMULAE

1. $\text{Gain} = (\text{S.P.}) - (\text{C.P.})$

2. $\text{Loss} = (\text{C.P.}) - (\text{S.P.})$

3. Loss or gain is always reckoned on C.P.

4. Gain Percentage: (Gain %)

$$\text{Gain \%} = \left(\frac{\text{Gain} \times 100}{\text{C.P.}} \right)$$

5. Loss Percentage: (Loss %)

$$\text{Loss \%} = \left(\frac{\text{Loss} \times 100}{\text{C.P.}} \right)$$

6. Selling Price: (S.P.)

$$\text{SP} = \left[\frac{(100 + \text{Gain \%})}{100} \times \text{C.P.} \right]$$

7. Selling Price: (S.P.)

$$SP = \left[\frac{(100 - \text{Loss \%})}{100} \times \text{C.P.} \right]$$

8. Cost Price: (C.P.)

$$\text{C.P.} = \left[\frac{100}{(100 + \text{Gain \%})} \times \text{S.P.} \right]$$

9. Cost Price: (C.P.)

$$\text{C.P.} = \left[\frac{100}{(100 - \text{Loss \%})} \times \text{S.P.} \right]$$

10. If an article is sold at a gain of say 35%, then S.P. = 135% of C.P.

11. If an article is sold at a loss of say, 35% then S.P. = 65% of C.P.

12. When a person sells two similar items, one at a gain of say $x\%$, and the other at a loss of $x\%$, then the seller always incurs a loss given by:

$$\text{Loss \%} = \left(\frac{\text{Common Loss and Gain \%}}{10} \right)^2 = \left(\frac{x}{10} \right)^2.$$

13. If a trader professes to sell his goods at cost price, but uses false weights, then

$$\text{Gain \%} = \left[\frac{\text{Error}}{(\text{True Value}) - (\text{Error})} \times 100 \right] \%.$$

Profit and Loss - General Questions

1.

Alfred buys an old scooter for Rs. 4700 and spends Rs. 800 on its repairs. If he sells the scooter for Rs. 5800, his gain percent is:

4	4	%
	7	

5	5	%
	11	

10%

12%

2. The cost price of 20 articles is the same as the selling price of x articles. If the profit is 25%, then the value of x is:

15

16

18

25

3. If selling price is doubled, the profit triples. Find the profit percent.

66	2
	3

100

105	1
-----	---

120

4. In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?

30%

70%

100%

250%

5. A vendor bought toffees at 6 for a rupee. How many for a rupee must he sell to gain 20%?

3

4

5

6

6. The percentage profit earned by selling an article for Rs. 1920 is equal to the percentage loss incurred by selling the same article for Rs. 1280. At what price should the article be sold to make 25% profit?

Rs. 2000

Rs. 2200

Rs. 2400

Data inadequate

7. A shopkeeper expects a gain of 22.5% on his cost price. If in a week, his sale was of Rs. 392, what was his profit?

Rs. 18.20

Rs. 70

Rs. 72

Rs. 88.25

8. A man buys a cycle for Rs. 1400 and sells it at a loss of 15%. What is the selling price of the cycle?

Rs. 1090

Rs. 1160

Rs. 1190

Rs. 1202

9. Sam purchased 20 dozens of toys at the rate of Rs. 375 per dozen. He sold each one of them at the rate of Rs. 33. What was his percentage profit?

3.5

4.5

5.6

6.5

10. Some articles were bought at 6 articles for Rs. 5 and sold at 5 articles for Rs. 6. Gain percent is:

30%

33	1	%
	3	

35%

44%

11.

On selling 17 balls at Rs. 720, there is a loss equal to the cost price of 5 balls. The cost price of a ball is:

Rs. 45

Rs. 50

Rs. 55

Rs. 60

12. When a plot is sold for Rs. 18,700, the owner loses 15%. At what price must that plot be sold in order to gain 15%?

Rs. 21,000

Rs. 22,500

Rs. 25,300

Rs. 25,800

13. 100 oranges are bought at the rate of Rs. 350 and sold at the rate of Rs. 48 per dozen. The percentage of profit or loss is:

14	$\frac{2}{7}$	% gain
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15% gain

14	$\frac{2}{7}$	% loss
----	---------------	--------

15 % loss

14. A shopkeeper sells one transistor for Rs. 840 at a gain of 20% and another for Rs. 960 at a loss of 4%. His total gain or loss percent is:

5	$\frac{15}{17}$	% loss
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5	$\frac{15}{17}$	% gain
---	-----------------	--------

6	$\frac{2}{3}$	% gain
---	---------------	--------

None of these

15. A trader mixes 26 kg of rice at Rs. 20 per kg with 30 kg of rice of other variety at Rs. 36 per kg and sells the mixture at Rs. 30 per kg. His profit percent is:

No profit, no loss

5%

8%

10%

None of these

Profit and Loss - Data Sufficiency 1

1.

A shopkeeper sells some articles at the profit of 25% on the original price. What is the exact amount of profit? To find the answer, which of the following information given in Statements I

	Sale price of the article
	Number of articles sold

Only I is necessary

Only II is necessary

Either I or II is necessary

Both I and II are necessary

None of these

2.

A shopkeeper sells some toys at Rs. 250 each. What percent profit does he make? To find the answer, which of the following

	Number of toys sold.
	Cost price of each toy.

Only I is necessary

Only II is necessary

Both I and II are necessary

Either I or II is necessary

None of these

Profit and Loss - Data Sufficiency 2

Directions to Solve

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question. Read the both statements and

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1.

A man mixes two types of rice (X and Y) and sells the mixture at the rate of Rs. 17 per kg. Find his profit percentage.

The rate of X is Rs. 20 per kg.

The rate of Y is Rs. 13 per kg.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

2.

By selling a product with 20% profit, how much profit was earned?

I. The difference between cost and selling price is Rs. 40.

II. The selling price is 120 percent of the cost price.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

Profit and Loss - Data Sufficiency 3

Directions to Solve

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

1.

By selling an article what is the profit percent gained?	
I.	5% discount is given on list price.
II.	If discount is not given, 20% profit is gained.
III.	The cost price of the articles is Rs. 5000.

Only I and II

Only II and III

Only I and III

All I, II and III

None of these

2.

What was the percentage of discount given?	
I.	23.5% profit was earned by selling an almirah for Rs. 12,350.
II.	If there were no discount, the earned profit would have been 30%
III.	The cost price of the almirah was Rs. 10,000.

Only I and II

Only II and III

Only I and III

Any two of the three

None of these

Profit and Loss - Data Sufficiency 4

Directions to Solve

Each of these questions is followed by three statements. You have to study the question and all the three statements given to decide whether any information provided in the statement(s) is redundant and can be dispensed with while answering the given question.

1.

What is the percent profit earned by the shopkeeper on selling the	
	Labeled price of the articles sold was 130% of the cost price.
I	Cost price of each article was Rs. 550.
II	A discount of 10% on labeled price was offered.

Only I

Only II

I and III

All the three are required

Question cannot be answer even with information in all the three statements.

Partnership - Formulas

Partnership:

When two or more than two persons run a business jointly, they are called partners and the deal is known as partnership.

Ratio of Divisions of Gains:

When investments of all the partners are for the same time, the gain or loss is distributed among the partners in the ratio of their investments.

Suppose A and B invest Rs. x and Rs. y respectively for a year in a business, then at the end of the year:

$$(\text{A's share of profit}) : (\text{B's share of profit}) = x : y.$$

When investments are for different time periods, then equivalent capitals are calculated for a unit of time by taking (capital $\times$ number of units of time). Now gain or loss is divided in the ratio of these capitals.

Suppose A invests Rs. x for p months and B invests Rs. y for q months then,

$$(\text{A's share of profit}) : (\text{B's share of profit}) = xp : yq.$$

Working and Sleeping Partners:

A partner who manages the the business is known as a working partner and the one who simply invests the money is a sleeping partner.

Partnership - General Questions

1. A and B invest in a business in the ratio 3 : 2. If 5% of the total profit goes to charity and A's share is Rs. 855, the total profit is:

Rs. 1425

Rs. 1500

Rs. 1537.50

Rs. 1576

2. A, B and C jointly thought of engaging themselves in a business venture. It was agreed that A would invest Rs. 6500 for 6 months, B, Rs. 8400 for 5 months and C, Rs. 10,000 for 3 months. A wants to be the working member for which, he was to receive 5% of the profits. The profit earned was Rs. 7400. Calculate the share of B in the profit.

Rs. 1900

Rs. 2660

Rs. 2800

Rs. 2840

3. A, B and C enter into a partnership in the ratio $\frac{7}{2} : \frac{4}{3} : \frac{6}{5}$. After 4 months, A increases his share 50%. If the total profit at the end of one year be Rs. 21,600, then B's share in the profit is:

Rs. 2100

Rs. 2400

Rs. 3600

Rs. 4000

4. A, B, C subscribe Rs. 50,000 for a business. A subscribes Rs. 4000 more than B and B Rs. 5000 more than C. Out of a total profit of Rs. 35,000, A receives:

Rs. 8400

Rs. 11,900

Rs. 13,600

Rs. 14,700

5. Three partners shared the profit in a business in the ratio 5 : 7 : 8. They had partnered for 14 months, 8 months and 7 months respectively. What was the ratio of their investments?

5 : 7 : 8

20 : 49 : 64

38 : 28 : 21

None of these

6. A starts business with Rs. 3500 and after 5 months, B joins with A as his partner. After a year, the profit is divided in the ratio 2 : 3. What is B's contribution in the capital?

Rs. 7500

Rs. 8000

Rs. 8500

Rs. 9000

7. A and B entered into partnership with capitals in the ratio 4 : 5.

After 3 months, A withdrew $\frac{1}{4}$ of his capital and B withdrew $\frac{1}{5}$ of his capital. The gain at the end of 10 months was Rs. 760. A's share in this profit is:

Rs. 330

Rs. 360

Rs. 380

Rs. 430

8. A and B started a partnership business investing some amount in the ratio of 3 : 5. C joined then after six months with an amount equal to that of B. In what proportion should the profit at the end of one year be distributed among A, B and C?

3 : 5 : 2

3 : 5 : 5

6 : 10 : 5

Data inadequate

9. A, B, C rent a pasture. A puts 10 oxen for 7 months, B puts 12 oxen for 5 months and C puts 15 oxen for 3 months for grazing. If the rent of the pasture is Rs. 175, how much must C pay as his share of rent?

Rs. 45

Rs. 50

Rs. 55

Rs. 60

10. A and B started a business in partnership investing Rs. 20,000 and Rs. 15,000 respectively. After six months, C joined them with Rs. 20,000. What will be B's share in total profit of Rs. 25,000 earned at the end of 2 years from the starting of the business?

Rs. 7500

Rs. 9000

Rs. 9500

Rs. 10,000

11. A began a business with Rs. 85,000. He was joined afterwards by B with Rs. 42,500. For how much period does B join, if the profits at the end of the year are divided in the ratio of 3 : 1?

4 months

5 months

6 months

8 months

12. Aman started a business investing Rs. 70,000. Rakhi joined him after six months with an amount of Rs. 1,05,000 and Sagar joined them with Rs. 1.4 lakhs after another six months. The amount of profit earned should be distributed in what ratio among Aman, Rakhi and Sagar respectively, 3 years after Aman started the business?

7 : 6 : 10

12 : 15 : 16

42 : 45 : 56

Cannot be determined

13. Arun, Kamal and Vinay invested Rs. 8000, Rs. 4000 and Rs. 8000 respectively in a business. Arun left after six months. If after eight months, there was a gain of Rs. 4005, then what will be the share of Kamal?

Rs. 890

Rs. 1335

Rs. 1602

Rs. 1780

14. Simran started a software business by investing Rs. 50,000. After six months, Nanda joined her with a capital of Rs. 80,000. After 3 years, they earned a profit of Rs. 24,500. What was Simran's share in the profit?

Rs. 9,423

Rs. 10,250

Rs. 12,500

Rs. 10,500

Partnership - Data Sufficiency 1

Directions to Solve

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1.

Ravi, Gagan and Nitin are running a business firm in partnership. What is Gagan's share in the profit earned by them?	
	I Ravi, Gagan and Nitin invested the amounts in the ratio of 2 : 4 :
	II Nitin's share in the profit is Rs. 8750.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

2.

Rahul, Anurag and Vivek started a business together. In what proportion would the annual profit be distributed among them?	
	Rahul got one-fourth of the profit.
I	Rahul and Vivek contributed 75% of the total investment.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

Partnership - Data Sufficiency 2

Directions to Solve

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

1.

How much did Rohit get as profit at the year-end in the business done by Nitin, Rohit and Kunal?	
I	Kunal invested Rs. 8000 for nine months, his profit was $\frac{3}{2}$ times that of Rohit's and his investment was four times that of Nitin.
II	Nitin and Rohit invested for one year in the proportion 1 : 2 respectively.
III	The three together got Rs. 1000 as profit at the year end.

Only I and II

Only I and III

Question cannot be answered even with the information in all the three statements.

All I, II and III

None of these

2.

What is R's share of profit in a joint venture?	
I	Q started business investing Rs. 80,000.
II	R joined him after 3 months.
III	P joined after 4 months with a capital of Rs. 1,20,000 and got Rs

All I, II and III

I and III only

II and III only

Even with all I, II and III, the answer cannot be arrived at

None of these

Partnership - Data Sufficiency 3

Directions to Solve

Each of these questions is followed by three statements. You have to study the question and all the three statements given to decide whether any information provided in the statement(s) is redundant and can be dispensed with while answering the given question.

1.

Three friends, P, Q and R started a partnership business investing money in the ratio of 5 : 4 : 2 respectively for a period of 3 years.

Total amount invested in the business in Rs. 22,000.

I Profit earned at the end of 3 years is $\frac{3}{8}$ of the total investment.

II The average amount of profit earned per year is Rs. 2750.

I or II or III

Either III only, or I and II together

Any two of the three

All I, II and III are required.

None of these

Percentage - Formulas

1. Concept of Percentage:

By a certain percent, we mean that many hundredths.

Thus, x percent means x hundredths, written as $x\%$.

To express $x\%$ as a fraction: We have, $x\% = \frac{x}{100}$.

$$\text{Thus, } 20\% = \frac{20}{100} = \frac{1}{5}.$$

To express $\frac{a}{b}$ as a percent: We have, $\frac{a}{b} = \left(\frac{a}{b} \times 100 \right)\%$.

$$\text{Thus, } \frac{1}{4} = \left(\frac{1}{4} \times 100 \right)\% = 25\%.$$

2. Percentage Increase/Decrease:

If the price of a commodity increases by $R\%$, then the reduction in consumption so as not to increase the expenditure is:

$$\left[\frac{R}{(100 + R)} \times 100 \right]\%$$

If the price of a commodity decreases by $R\%$, then the increase in consumption so as not to decrease the expenditure is:

$$\left[\frac{R}{(100 - R)} \times 100 \right]\%$$

3. Results on Population:

Let the population of a town be P now and suppose it increases at the rate of $R\%$ per annum, then:

1. Population after n years $= P \left(1 + \frac{R}{100} \right)^n$

2. Population n years ago $= \frac{P}{\left(1 + \frac{R}{100} \right)^n}$

4. Results on Depreciation:

Let the present value of a machine be P . Suppose it depreciates at the rate of $R\%$ per annum. Then:

1. Value of the machine after n years $= P \left(1 - \frac{R}{100} \right)^n$

2. Value of the machine n years ago $= \frac{P}{\left(1 - \frac{R}{100} \right)^n}$

3. If A is $R\%$ more than B , then B is less than A by $\left[\frac{R}{(100 + R)} \times 100 \right] \%$.

4. If A is $R\%$ less than B , then B is more than A by $\left[\frac{R}{(100 - R)} \times 100 \right] \%$.

Percentage - General Questions

1.

A batsman scored 110 runs which included 3 boundaries and 8 sixes. What percent of his total score did he make by running between the wickets?

45%

45	5	%
	1	

54	6	%
	1	

55%

2. Two students appeared at an examination. One of them secured 9 marks more than the other and his marks was 56% of the sum of their marks. The marks obtained by them are:

39, 30

41, 32

42, 33

43, 34

3. A fruit seller had some apples. He sells 40% apples and still has 420 apples. Originally, he had:

588 apples

600 apples

672 apples

700 apples

4. What percentage of numbers from 1 to 70 have 1 or 9 in the unit's digit?

1

14

20

21

5. If $A = x\%$ of y and $B = y\%$ of x , then which of the following is true?

A is smaller than B.

A is greater than B

Relationship between A and B cannot be determined.

If x is smaller than y , then A is greater than B.

None of these

6. If 20% of $a = b$, then $b\%$ of 20 is the same as:

4% of a

5% of a

20% of a

None of these

7. In a certain school, 20% of students are below 8 years of age.

The number of students above 8 years of age is $\frac{2}{3}$ of the number of students of 8 years of age which is 48. What is the total number of students in the school?

72

80

120

150

100

8. Two numbers A and B are such that the sum of 5% of A and 4% of B is two-third of the sum of 6% of A and 8% of B. Find the ratio of A : B.

2 : 3

1 : 1

3 : 4

4 : 3

9.

A student multiplied a number by	$\frac{3}{5}$	instead of	$\frac{5}{3}$	.
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What is the percentage error in the calculation?

34%

44%

54%

64%

10. In an election between two candidates, one got 55% of the total valid votes, 20% of the votes were invalid. If the total number of votes was 7500, the number of valid votes that the other candidate got, was:

2700

2900

3000

3100

11. Three candidates contested an election and received 1136, 7636 and 11628 votes respectively. What percentage of the total votes did the winning candidate get?

57%

60%

65%

90%

12. Two tailors X and Y are paid a total of Rs. 550 per week by their employer. If X is paid 120 percent of the sum paid to Y, how much is Y paid per week?

Rs. 200

Rs. 250

Rs. 300

None of these

13. Gauri went to the stationers and bought things worth Rs. 25, out of which 30 paise went on sales tax on taxable purchases. If the tax rate was 6%, then what was the cost of the tax free items?

Rs. 15

Rs. 15.70

Rs. 19.70

Rs. 20

14. Rajeev buys good worth Rs. 6650. He gets a rebate of 6% on it. After getting the rebate, he pays sales tax @ 10%. Find the amount he will have to pay for the goods.

Rs. 6876.10

Rs. 6999.20

Rs. 6654

Rs. 7000

15. The population of a town increased from 1,75,000 to 2,62,500 in a decade. The average percent increase of population per year is:

4.37%

5%

6%

8.75%

Problems on Ages - Formulas

Important Formulas on "Problems on Ages" :

1. If the current age is x , then n times the age is nx .
2. If the current age is x , then age n years later/hence = $x + n$.
3. If the current age is x , then age n years ago = $x - n$.
4. The ages in a ratio $a : b$ will be ax and bx .

5. If the current age is x , then	1	of the age is	x
	n		n

Problems on Ages - General Questions

1. Father is aged three times more than his son Ronit. After 8 years, he would be two and a half times of Ronit's age. After further 8 years, how many times would he be of Ronit's age?

2 times

2	1	times
	2	

2	3	times
	4	

3 times

2. The sum of ages of 5 children born at the intervals of 3 years each is 50 years. What is the age of the youngest child?

4 years

8 years

10 years

None of these

3. A father said to his son, "I was as old as you are at the present at the time of your birth". If the father's age is 38 years now, the son's age five years back was:

14 years

19 years

33 years

38 years

4. A is two years older than B who is twice as old as C. If the total of the ages of A, B and C be 27, then how old is B?

7

8

9

10

11

5. Present ages of Sameer and Anand are in the ratio of 5 : 4 respectively. Three years hence, the ratio of their ages will become 11 : 9 respectively. What is Anand's present age in years?

24

27

40

Cannot be determined

None of these

6. A man is 24 years older than his son. In two years, his age will be twice the age of his son. The present age of his son is:

14 years

18 years

20 years

22 years

7. Six years ago, the ratio of the ages of Kunal and Sagar was 6 : 5. Four years hence, the ratio of their ages will be 11 : 10. What is Sagar's age at present?

16 years

18 years

20 years

Cannot be determined

None of these

8. The sum of the present ages of a father and his son is 60 years. Six years ago, father's age was five times the age of the son. After 6 years, son's age will be:

12 years

14 years

18 years

20 years

9. At present, the ratio between the ages of Arun and Deepak is 4 : 3. After 6 years, Arun's age will be 26 years. What is the age of Deepak at present ?

12 years

15 years

19 and half

21 years

10. Sachin is younger than Rahul by 7 years. If their ages are in the respective ratio of 7 : 9, how old is Sachin?

16 years

18 years

28 years

24.5 years

None of these

11. The present ages of three persons in proportions 4 : 7 : 9. Eight years ago, the sum of their ages was 56. Find their present ages (in years).

8, 20, 28

16, 28, 36

20, 35, 45

None of these

12. Ayesha's father was 38 years of age when she was born while her mother was 36 years old when her brother four years younger to her was born. What is the difference between the ages of her parents?

2 years

4 years

6 years

8 years

13. A person's present age is two-fifth of the age of his mother. After 8 years, he will be one-half of the age of his mother. How old is the mother at present?

32 years

36 years

40 years

48 years

14. Q is as much younger than R as he is older than T. If the sum of the ages of R and T is 50 years, what is definitely the difference between R and Q's age?

1 year

2 years

25 years

Data inadequate

None of these

15. The age of father 10 years ago was thrice the age of his son. Ten years hence, father's age will be twice that of his son. The ratio of their present ages is:

5 : 2

7 : 3

9 : 2

13 : 4

Problems on Ages - Data Sufficiency 1

Directions to Solve

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1.

What is Sonia's present age?	
I.	Sonia's present age is five times Deepak's present age.
II.	Five years ago her age was twenty-five times Deepak's age at th

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

2.

Average age of employees working in a department is 30 years. In the next year, ten workers will retire. What will be the average age	
	Retirement age is 60 years.
	There are 50 employees in the department.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

3.

Divya is twice as old as Shruti. What is the difference in their	
I.	Five years hence, the ratio of their ages would be 9 : 5.
II.	Ten years back, the ratio of their ages was 3 : 1.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

Problems on Ages - Data Sufficiency 2

Directions to Solve

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

1.

What is Arun's present age?
I Five years ago, Arun's age was double that of his son's age at that time.
II Present ages of Arun and his son are in the ratio of 11 : 6 respectively.
III Five years hence, the respective ratio of Arun's age and his son's age will be 13 : 6.

Only I and II

Only II and III

Only I and III

Any two of the three

None of these

2.

What is Ravi's present age?
I The present age of Ravi is half of that of his father.
II After 5 years, the ratio of Ravi's age to that of his father's age will be 1 : 4.
III Ravi is 5 years younger than his brother.

I and II only

II and III only

I and III only

All I, II and III

Even with all the three statements answer cannot be determined.

3.

What is the present age of Tanya?	
I	The ratio between the present ages of Tanya and her brother Rahul is 3:4.
II	After 5 years the ratio between the ages of Tanya and Rahul will be 4:5.
III	Rahul is 5 years older than Tanya.

I and II only

II and III only

I and III only

All I, II and III

Any two of the three

Problems on Ages - Data Sufficiency 3

Directions to Solve

Each of these questions is followed by three statements. You have to study the question and all the three statements given to decide whether any information provided in the statement(s) is redundant and can be dispensed with while answering the given question.

1.

What will be the ratio between ages of Sam and Albert after 5	
I.	Sam's present age is more than Albert's present age by 4 years
II.	Albert's present age is 20 years.
III.	The ratio of Albert's present age to Sam's present age is 5 : 6.

Any two of I, II and III

II only

III only

I or III only

II or III only

2.

What is the difference between the present ages of Ayush and	
I	The ratio between Ayush's present age and his age after 8 years
II	The ratio between the present ages of Ayush and Deepak is 4 : 3
III	The ratio between Deepak's present age and his age four years

Any two of I, II and III

I or III only

Any one of the three

All I, II and III are required

Even with all I, II and III, the answer cannot be obtained.